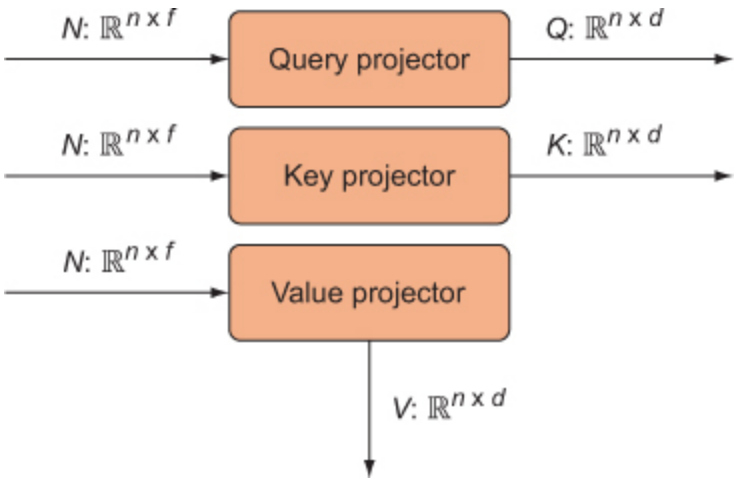
still achieve good performance and interpretability with the absolute encoding.

We then pass this initial node matrix through three different linear layers to project it into three different matrices with a potentially different channel dimension (which we will call *node-feature dimension* from this point), as shown in figure 10.15.

Figure 10.15. The projection step in self-attention. The input nodes are projected into a (usually) higher-dimensional feature space by simple matrix multiplication.



Once we multiply the query and key matrices, we get an unnormalized attention weight matrix, A:$b \times n \times n$, where b = batch and n = the number of nodes. We then normalize it by applying softmax across the rows (dimension 1, counting from 0) such that each row sums to 1. This forces each node to only pay attention to a small number of other nodes, or to spread its attention very thinly across many nodes.

Then we multiply the attention matrix by the value matrix to get an updated node matrix, such that each node is now a weighted combination of all the other nodes. So if node 0 pays strong attention to nodes 5 and 9 but ignores the others, once we multiply the attention matrix with the value matrix, node 0 will be updated to be a weighted combination of nodes 5 and 9 (and itself, because nodes generally pay some attention to themselves). This general operation is termed *message passing* because each node sends a message (i.e., its own feature vector) to the nodes to which it is connected.